

Mental Ray Lights

Directional, point, and spotlights in 3ds Max can be used with the Mental Ray for 3ds Max renderer, and most 3ds Max lighting parameters translate with no effort. Each light has rollouts for the mental ray options, which control indirect illumination and light shaders. Indirect illumination in mental ray simulates the way light naturally moves through a scene. Global illumination (GI) simulates how light bounces off surfaces in a scene and can create very soft and realistic renderings. Caustics simulate the way light transmits through an object or reflects off it. Light shaders are custom plug-ins for mental ray that affect the light's behaviour for creating special effects.

Skylights are a special type of light that simulates natural diffuse lighting. This light is great for adding realism to a scene, but it does take longer to render. When using a skylight, the Light-Tracer must be turned on in the Render Scene dialog box.

Adjusting Lights

Lights can be manipulated by using the Modify panel or directly in a viewport. Lights can be aimed and positioned by using the Move and Rotate tools, and if a light has a target, the target can be moved as well. The Select And Manipulate tool can be used to adjust the cone angle and rotation of a spotlight or direct light.

4.3 Creating Shadows

Almost all objects cast shadows, and creating realistic shadows is one of the keys to truly simulating reality in 3ds Max. Shadows not only add realism, but contribute greatly to the mood of a scene; they help the eye determine the placement of objects in a scene, and they visually anchor objects to the ground.

Shadow Parameters

Parameters affecting all types of shadows can be adjusted with the Shadow Parameters rollout in the light's Modify panel. These parameters affect the colour of the shadow as well as the density of the shadow. You can also have atmospheric effects cast shadows.

Depth Map Shadows

Depth map shadows are used extensively by artists because they render quickly. Though not quite as precise as raytraced shadows, they're great for most general shadowing applications. 3ds Max creates a depth map shadow by rendering